

The software for NA62 is split into three main C++ packages: NA62MC, NA62Reconstruction and NA62Analysis. NA62MC is a GEANT4 based Monte Carlo simulation package that simulates the NA62 experiment including the detector geometry and the full simulation of kaon decays, used as a comparison to the data collected by the experiment. NA62Reconstruction is a package that is used to reconstruct either the Monte Carlo data or collected data from the experiment for all detectors. NA62Analysis is used by the physicists to analyse the data produced from the reconstruction to produce all the physics studies that are conducted at NA62.

The NA62 software was designed for use only with the NA62 experiment resulting in minimal configurability and flexibility, with the only option for users to change the software being to enable/disable sub-detectors. With the experiment foreseen to run up to CERN's Long shutdown 3 (LS3), an upgraded software package was needed to allow for the development of new detectors for the future kaon experiments. This software had to be flexible and configurable to allow for the simultaneous use of the software for NA62 data taking and physics analysis, as well as the development of new experiments.

0.1 FlexibleMC

0.1.1 Description

For the implementation of a flexible framework all detector classes were refactored to inherit from a common class which contained the basic functions and variables for all detectors allowing for all classes to be parsed using the same machinery.

The input to the simulation was then changed to allow for the configuration of all the detectors to be read in from a configuration file which contained any variables required for individual detectors and the X,Y,Z and rotation of the detectors. An example entry in the configuration file is shown in Figure 1 where a name was given for the station "STRAW0", along with chamber ID, chmaber type which defines the size of the straws and the chamber geometry which defines if the beam hole is symetric or not. The variables for each subdetector are defined in Table

Name	Arguments	X	Y	Z	θ_x	θ_y	θ_z
Spectrometer("STRAW0",	0, 0, 0)	0	0	183705.9	0	0	0

Figure 1: Example configuration for the STRAW0

Detector Name	Variables
STRAW	Chamber ID: 0,1,2,3 Chamber Type: 0(10mm), 1(5mm) Chamber Mode: 0(default), 1(symmetric)
Spectrometer Magnet	Magnetic field intensity
Vessel	Length InnerRadius OuterRadius Type("BlueTube", "BlueTubeNoRails", "GreyTube")

Table 1

When built inside the simulation, each detector was now contained within its own separate volume called the "Responsibility Region". This was a simple task for most detectors apart from the STRAWs and LAV which were made up of multiple

stations that were not continuous, resulting in multiple responsibility regions for each station.

0.1.2 Testing: Geometry and Magnetic Fields

After the simulation was upgraded all the geometries of each detector had to be validated that they were still correct with respect to the previous version. This was due to the complex nature of refactoring the detectors and that a number volumes had been changed. The validation was completed using the GEANT4 , which is a virtual particle present in the simulation package that will travel in a straight line and interact with all volumes present in the simulation. It was then possible to produce a large square beam of these particles and save every time the particle passed through a boundary between two volumes. Using this it was possible to check the placement and geometry of each detector to mm precision to ensure that the simulation was still accurate. An example of the geantino hits on a single view of STRAW chamber 1 is shown in Figure 2 where it is possible to see the features of the view. For the full validation a larger sample of Geantinos is used to get the required precision, however when there was a discrepancy they were sent in a much smaller area.

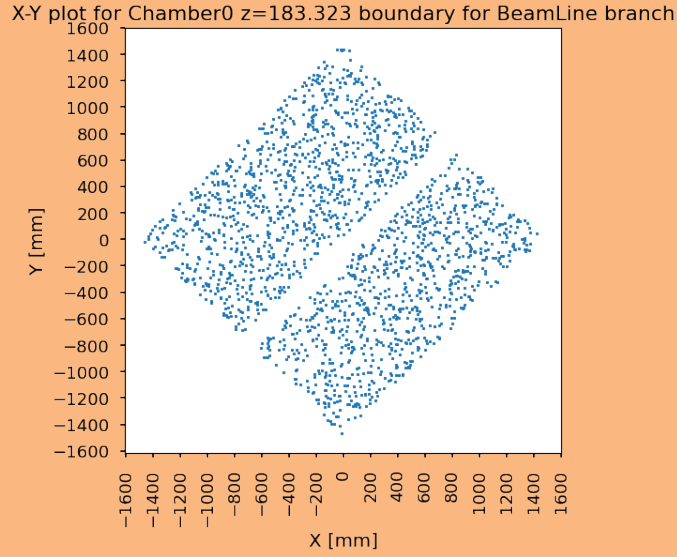
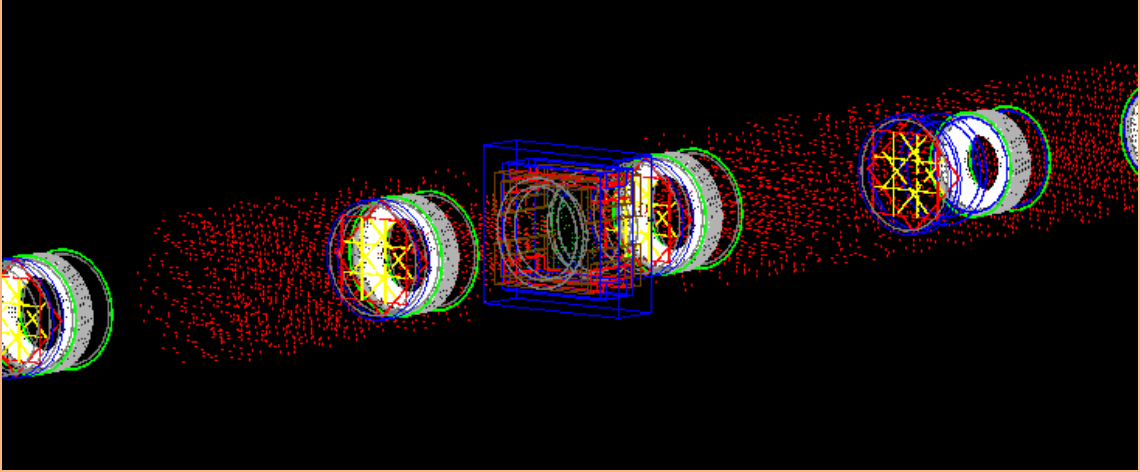


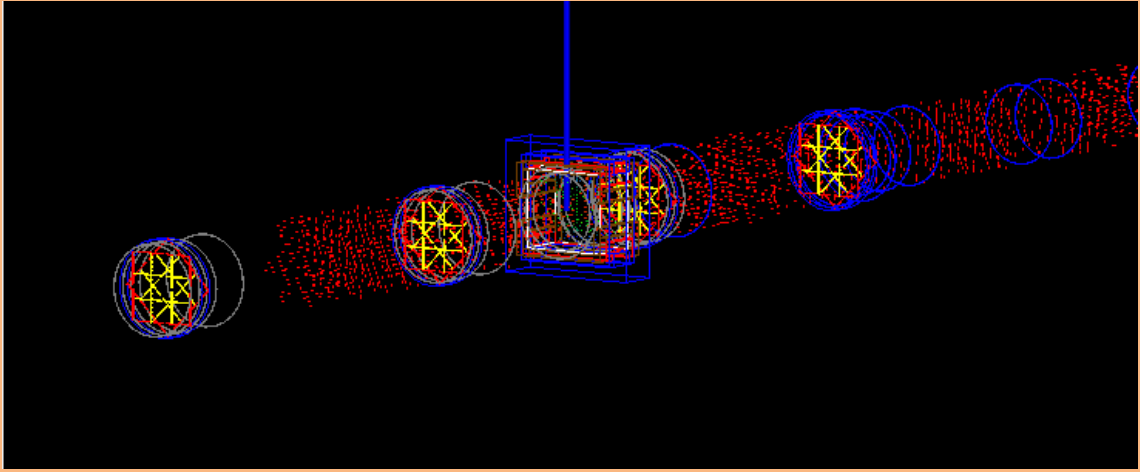
Figure 2: Geantino hits on a single view in straw chamber 1

To reduce the chance of overlaps and to clean up the simulation, a campaign to reduce the size of the unused space in all sub-detectors. The outer volume (Responsibility Region) for all detectors were reduced to just slightly larger than the physical detector which is possible to see in figure 3, as the magnetic fields are calculated in the all volumes not the world volume. In 3(a) the field was calculated outside of the actual physical sub-detector due to the larger outer volume, whereas in 3(b) the volume has been reduced to a few mm's bigger than the actual physical detector and as a result the field is calculated in the correct region.

To improve the usability for users a geometry check is completed every time before the simulation is ran. Each outer detector volume is checked for any overlaps with other detectors in all three dimensions taking into account any rotations and the shape of the outer volumes, if a overlap is found the simulation does not start and an error is produced to the user describing where the overlap is and what detectors are involved.



(a) Previous NA62MC Magnetic field calculations



(b) Upgraded NA62MC Magnetic field calculations

Figure 3: Visualisation from NA62MC of the magnetic field in the spectrometer region

The magnetic field in the simulation is based on real-world measurements of the magnetic field at NA62 from hall probes [1], with this information saved in the form of data files which is taken in and used to calculate the magnetic field at various points in the simulation. In NA62MC the magnetic field is split into three distinct regions; Bluetube which is the magnetic field within the decay region which is used for small corrections, fringe field which are the edges of the MNP33 magnetic field

and the MNP33 field which is the main magnetic field defined in the simulation from the MNP33 magnet. In the previous version of the simulation the position of the magnetic fields were hard coded, however in the upgraded version it was possible to move the MNP33 magnet to be placed in any position, requiring the magnetic field to be calculated relative from that position. An example of this flexibility is shown in Figure 4 where the MNP33 magnet was moved from 198 m to 163 m in front of the straw chamber 1 and the magnetic field was recalculated from that point within the constructed volumes.

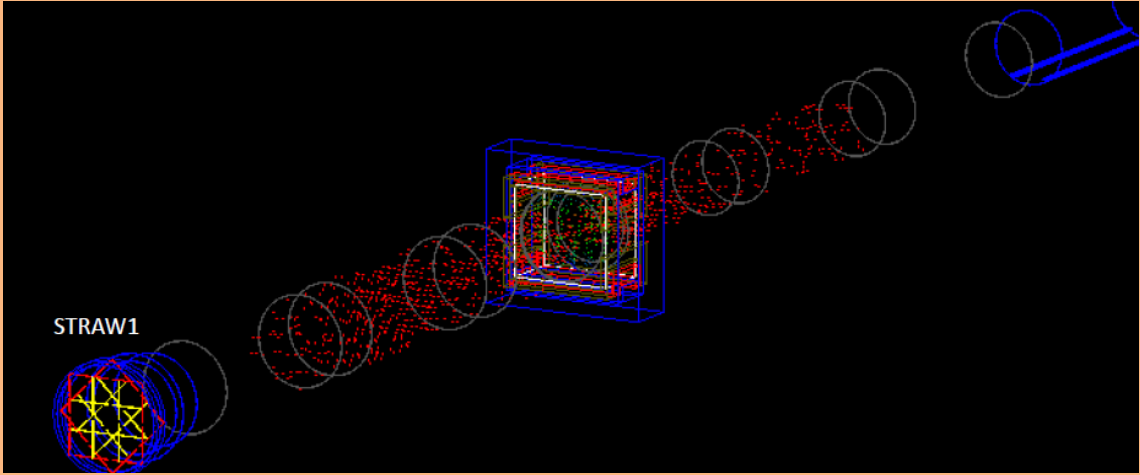


Figure 4: Visualisation from NA62MC with the MNP33 magnet moved in front of the straw chamber 1

The modification of the magnetic fields required validation, which was done using similar machinery to the geometry test completed above but with different GEANT particle, the charged Geantino. Similar to a normal Geantino, but they interact with magnetic fields allowing to ensure that the same tracks were being produced by the new version of the software. An example of this is shown in Figure 5 where it is possible to see the displacement of the charged Geantinos within the field regions. The exact XY positions of the hits would then be validated against a sample of

charged Geantinos with the exact mass, momentum and direction produced with the previous version of the software.

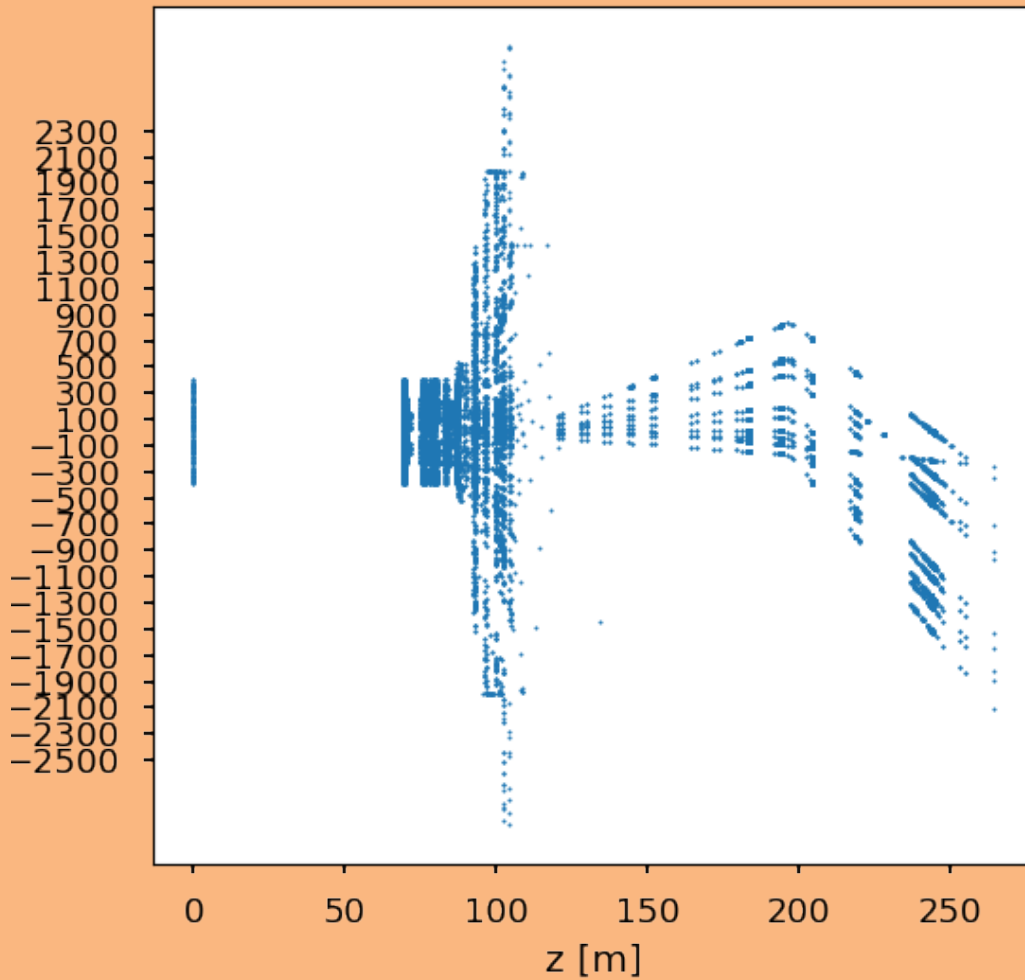


Figure 5: Charged Geantino hits on all detectors ADD EXTENT OF THE FIELD REGIONS

With minimal modification it was also possible to add the functionality to add multiple magnets in the simulation with their magnet fields producing a combined field. This was not implemented into the framework as its use was not foreseen but the machinery was developed and is ready for uses when required.

After the geometry and magnetic field tests were completed, full test productions of $K^+ \rightarrow \pi^+ \pi^0$, $K^+ \rightarrow \pi^+ \pi^- \pi^+$ and $K^+ \rightarrow \mu^+ \bar{\nu}$ were produced to test the validity of the existing NA62 setup along with the entire software chain (MC-;Reconstruction-;Analysis).

0.1.3 Testing: Symmetric NA62

It was also required to test the entire software chain using the new features of the flexible framework to ensure that when it was required and used in the future there were no issues. It was proposed to conduct a small study to investigate the feasibility of a symmetric NA62.

As described previously in Chapter ?? the current NA62 detector setup has been optimised for the search of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ with the current experiment set to run until CERN LS3. After LS3 it was foreseen that there would be the possibility of both charged and neutral kaon experiments located at CERN requiring a redesign of the NA62 detector setup. In the current NA62 setup, the beam is deflected in the positive x-direction by the TRIM5 magnet, which is located between GTK stations 2 and 3. The TRIM5 has a strength of $-0.3 Tm$, which equates to a 90 MeV kick, deflecting the beam to a maximum deflection of approximately +114 mm at straw chamber 2. After chamber 2, the MNP33 dipole magnet deflects the beam in the negative x direction with a momentum kick of 270 MeV. This requires the beam holes for the four Straw stations to be offset from the centre axis as shown in Figure 6 to allow for the undecayed charged beam to through the spectrometer. The mean beam centre of a charged Kaon at $75 GeV$ is shown by the red line starting from straw chamber 1 all the way to the LKr accounting for the interactions with the

TRIM5 and the MNP33 magnets.

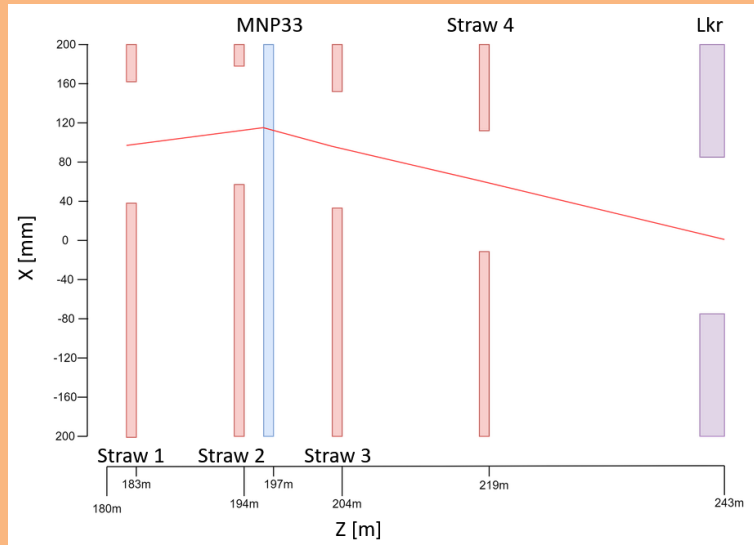


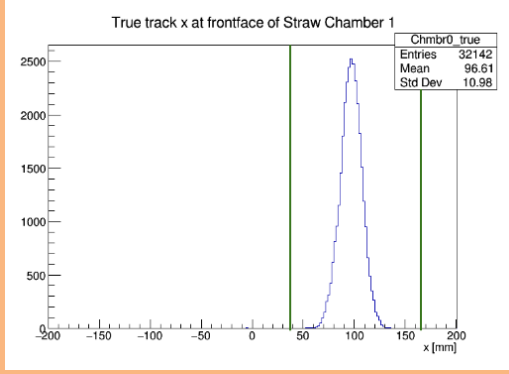
Figure 6: Diagram showing the path of the charged kaon beam, the straw chamber beam holes and the LKr beam pipe

The exact positions of the NA62 straw chambers and the beam holes are defined in Table 2.

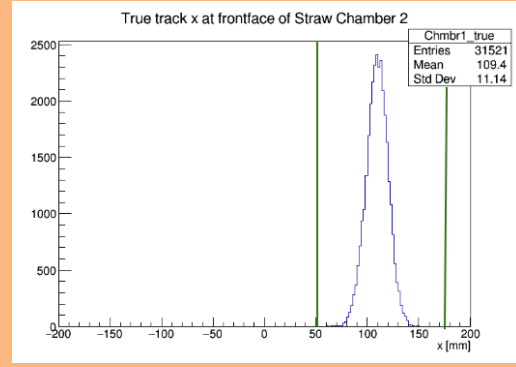
Straw Chamber	Z BackFace [m]	X Hole Displacement [mm]
1	183.7	101.2
2	194.26	114.4
3	204.64	92.4
4	219.07	52.8

Table 2: Straw chamber beam hole properties for NA62 detector setup

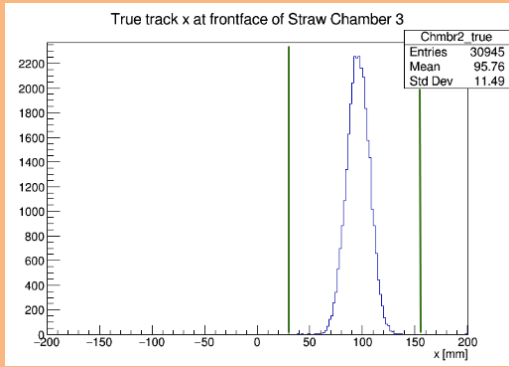
The beam profile at each Straw station is shown in Figure 7.



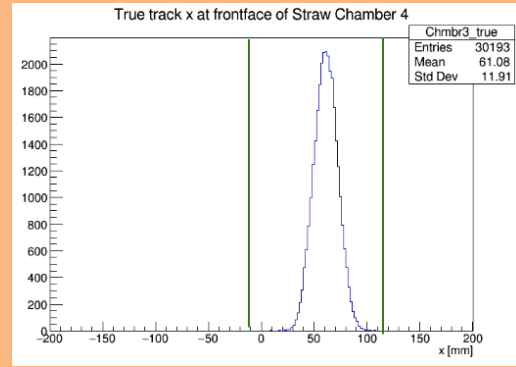
(a) Straw chamber 1



(b) Straw chamber 2



(c) Straw chamber 3



(d) Straw chamber 4

Figure 7: Plots of current NA62 beam coordinates at each straw chamber with the beam hole walls noted by the green vertical lines

To accommodate a neutral beam, the beam holes in the Straw stations would have to become centred at 0 in the X dimension, creating a symmetric detector design. However using the current NA62 beam setup it is clear the hole centred at 0 would be incompatible as from Figure 7 the beam would impact the detector. To fix this issue the TRIM5 magnet would be removed, allowing the beam to traverse along the z-axis until the interaction with the MNP33 magnetic field allowing the beam to pass through Straw chambers 1 and 2 but resulted in the beam impacting chamber 4 as shown in Figure 8.

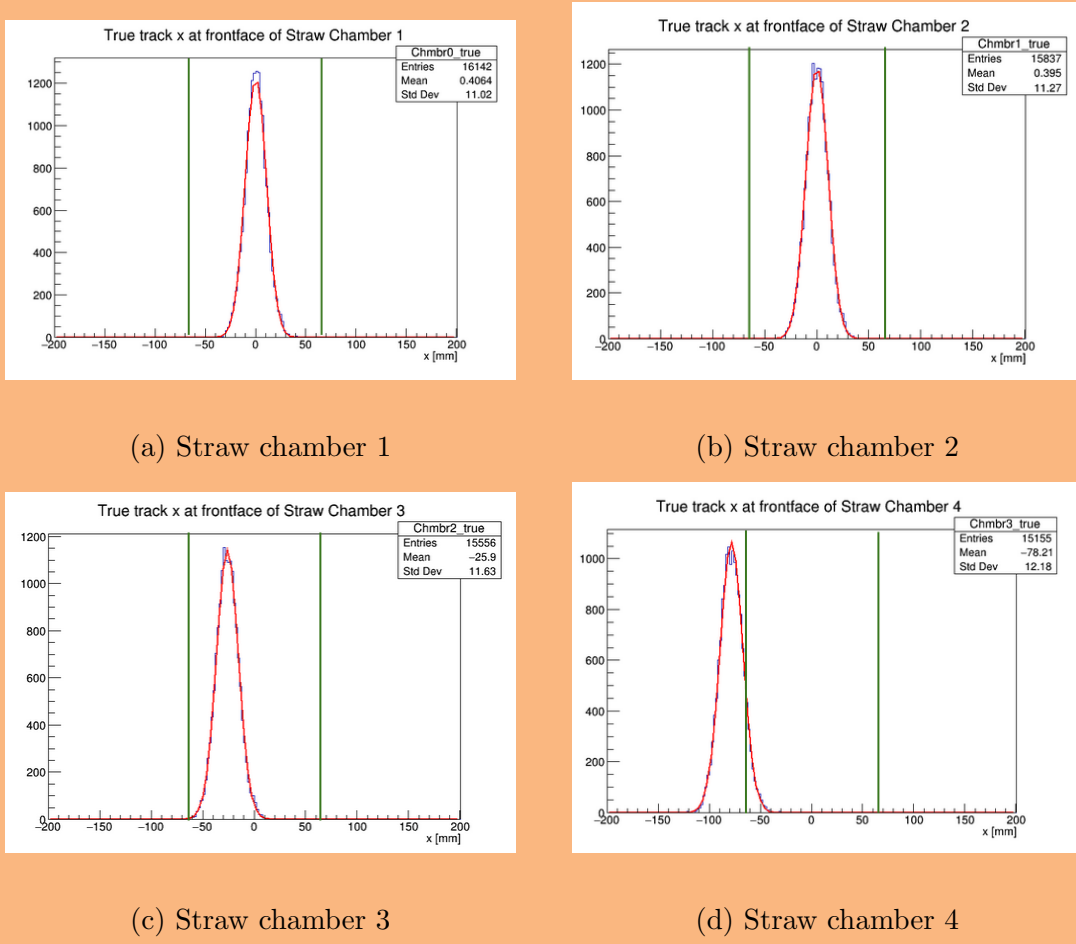


Figure 8: Plots of symmetric straw chambers with current NA62 beam parameters

In order to fit the beam within the last straw chamber three parameters from the NA62 could be modified:

1. Beam momentum (NA62 Nominal 75 GeV)
2. Chamber 4 positioning (NA62 Nominal back-face at 219.07 m)
3. MNP33 magnet field intensity (NA62 Nominal at $0.9 Tm$, which equates to a 270 MeV momentum kick)

To quantify if the beam fitted within the beam hole of the detector a safety parameter was defined:

$$\delta = \frac{x_0 - x_d}{\sigma} \quad (0.1)$$

Where x_0 is the mean of the fitted Gaussian of the beam at the detector plane, x_B is the x position of the beam hole wall (which was -63.8mm for the symmetric straw chambers), and σ is the sigma of the plotted Gaussian of the beam at the detector plane. The beam is defined to have fitted in the beam hole if $\delta \geq 3$.

The results of modifying the three parameters and their impact on the beam profile at straw chamber 4 is shown in Figure 9. Comparing results first for different momentum there is no clear change between 75 to 85 GeV due to the size of the beam increases as momentum increases counteracting the reduction in the deflected produced by the MNP33 magnet (MAYBE PUT BEAM SIGMA IN APPENDIX). Taking into account tht switching the beam momentum would cause other issues to the beam infrastructure it was decided to only continue with a 75 GeV beam (NA62 Nominal).

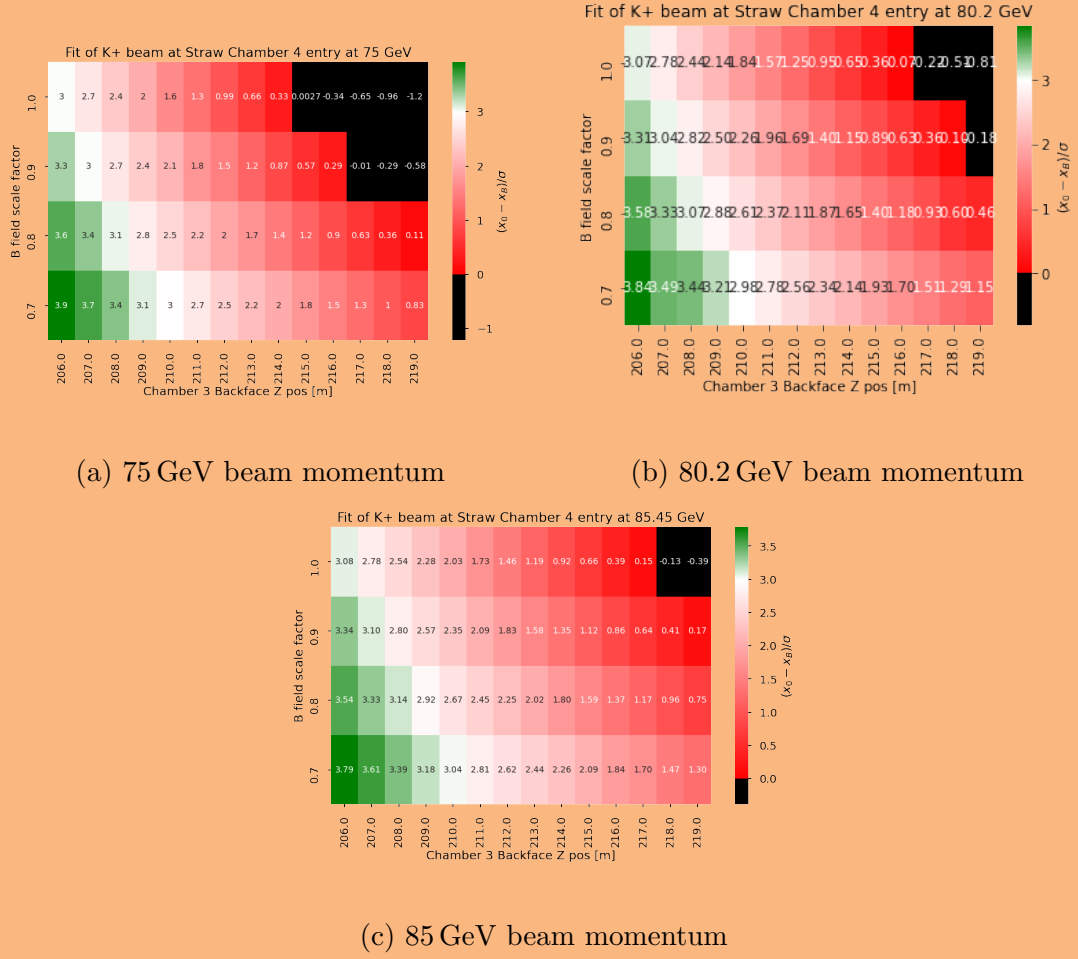


Figure 9: Plots of δ for different values of straw chamber 4 z position, MNP33 magnetic field and beam momentum

Focusing on Figure 9(a), only a small number of combinations produce a satisfactory safety factor. The optimal solution was with the MNP33 B field factored down by 0.7 and the straw station 4 moved to 206m, however this would place chamber 4 extremely close to chamber 3 which would impact the reconstruction resolution and would become apparent below. It was expected that a chamber position of $> 209\text{ m}$ would produce an acceptable resolution leaving few combinations available, other than ones at 0.7 MNP33 field scale. The setup with chamber 4 at 209m was chosen to ensure that the beam definitely fitted within the beam hole.

The next phase was to ensure that the beam would safely pass through the

beam hole through the LKr detector. This was imperative as unlike all most all other detectors which would be simple to move or slightly their positions, due to its size and position in the beam line it would make it difficult if not impossible to move. The Lkr as described in Chapter ?? is located 243m downstream from the target and is slightly offset in the x dimension by 4.9 mm. It has a beam pipe which passes through its centre with a radius of 80 mm and a thickness of 3 mm. As the MNP33 magnet deflects the beam in the negative x direction, the wall of the beam pipe of interest was located at -75.1 mm. It was noted that instead of moving the STRAW chamber 4 upstream by 10 m to 209 m, it would have be more beneficial to move the other chambers and the MNP33 magnet downstream towards chamber 4 by 10m instead, as shown in the Figure 10. However this produced results with the beam impacting the LKr detector at -84.49 mm. Not many options were left as the entire spectrometer could not be moved further downstream due the placement of the RICH, the distance between chamber 3 and 4 was already at its minimum and it was assumed that the distance between the MNP33 magnet and chamber 3 had previously been optimised.

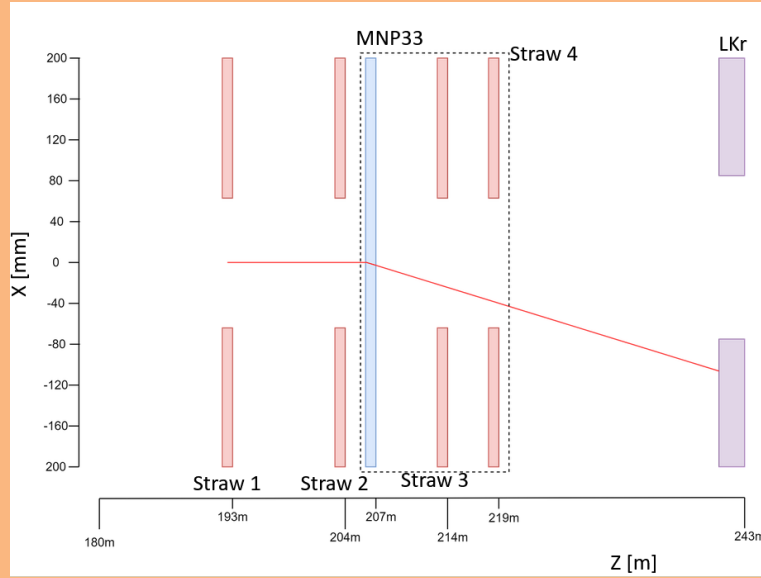


Figure 10: Diagram showing the updated setup with straw chambers 1,2,3 and the MNP33 magnet moving 10 m downstream

A solution was found that included utilising the previously removed TRIM5 magnet, displacing the charged kaon beam in the positive x direction before the MNP33 magnet. To allow for the biggest momentum kick possible straw chambers 1 and 2 were move back to their original positions, resulting in a modified setup shown in Figure 11.

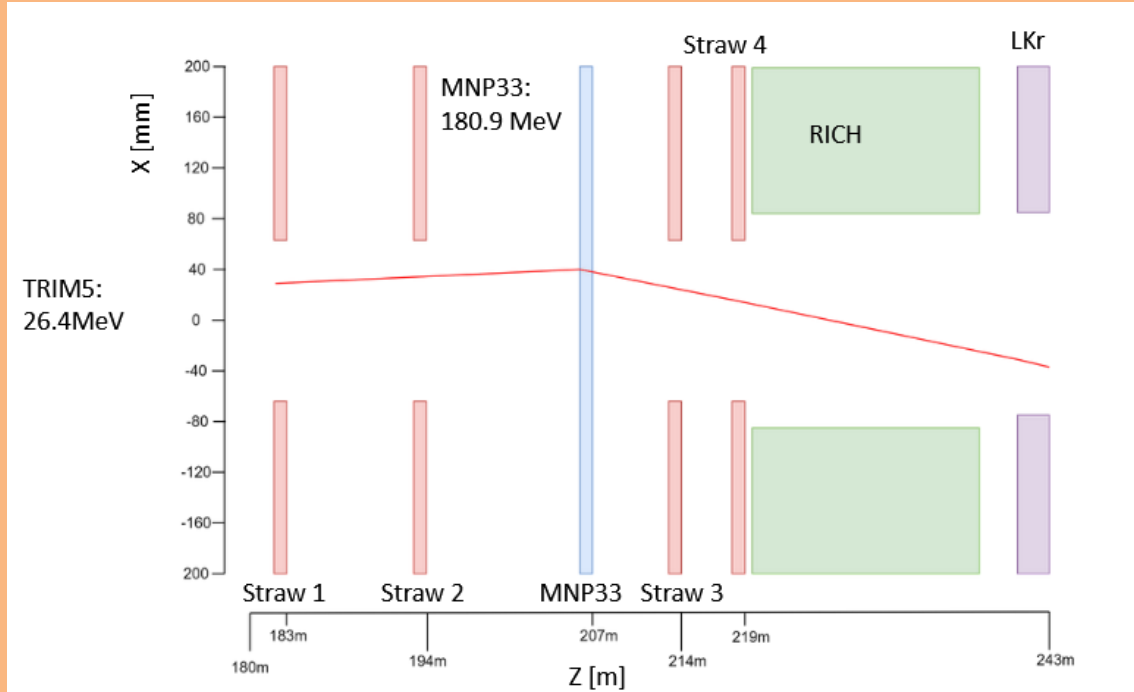


Figure 11: Diagram showing the proposed NA62 setup

The solution which provided the best beam placement in all straw chambers and the LKr and shown in Figure 12 was found when TRIM5 produced a -26.6 MeV kick (originally -90 MeV) and the MNP33 field scale was set at 0.67, producing a kick of 180.9 MeV (originally 270 MeV).

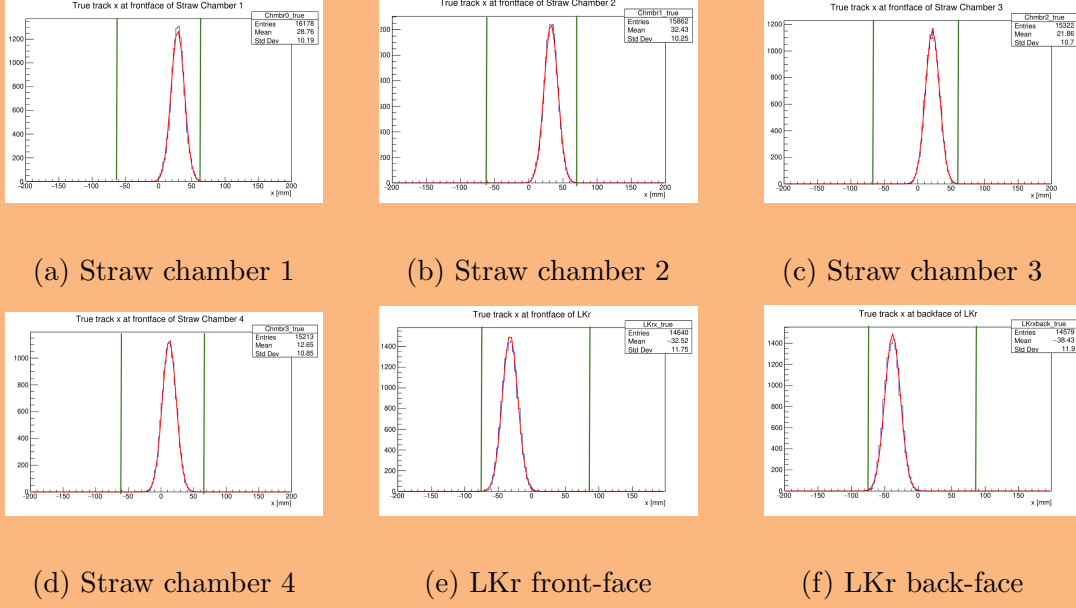


Figure 12: Plots of the beam profile in each of the straw chambers and the LKr front and back faces for the proposed symmetric setup

The overall safety margins for each Straw station and the LKr is provided in Table 3 with the following detector setup:

- MNP33 new z back-face position = 207.645 m (NA62 Nominal 197.645 m)
- MNP33 new field momentum kick = 180.9 MeV (NA62 Nominal 270 MeV)
- Chamber 3 new z back-face position = 214.645 m (NA62 Nominal 204.65 m)
- TRIM5 new field momentum kick = -26.6 MeV (NA62 Nominal -90 MeV)

Detector	Safety Margin	Number of beam particle outside of beam pipe/hole %
Chamber 1	3.46	0.035
Chamber 2	3.09	0.11
Chamber 3	3.99	0.025
Chamber 4	4.81	0.03
LKr Front	3.70	0.045
LKr Back	3.16	0.115

Table 3: Results showing the safety margin and number of original beam particles hit outside of the beam hole/pipe boundary for each chamber and the LKr

0.2 SlimMC

Bibliography

- [1] J. R. Fry, G. Ruggiero, and F. Bergsma, “Precision magnetic field mapping for CERN experiment NA62,” *Journal of Physics G: Nuclear and Particle Physics*, vol. 43, no. 12, p. 125004, nov 2016. [Online]. Available: <https://doi.org/10.1088/0954-3899/43/12/125004>